

Close the loop on invoice exceptions

A design exercise for a common utility finance scenario: *invoices come in above estimate, and the reasons don't get captured*. The design works as a new build or as an improvement to an existing app.

The problem

- 1 Invoices come in above estimate.** The reason lives in emails, calls, and memory.
- 2 The weekly review loses time to questioning numbers** instead of removing causes.
- 3 Financial controls need every variance tied to a reason code, an owner, and a decision.**

Power Apps | Invoice Exception Tracker

Invoice Exception Tracker Electric Ops · Finance Refreshed 9:12 AM JS

Exceptions queue
14 open · 3 need approval

Search invoice, vendor, or work order

Open Needs approval Resolved All

INV-20391 · Delta Line Constructors \$99,350 vs est. \$84,200 · WO-118427	+18.0% Open
INV-20388 · Sierra Civil & Boring \$41,900 vs est. \$36,750 · WO-118301	+14.0% Open
INV-20375 · Valley Vegetation Mgmt \$18,240 vs est. \$16,900 · WO-117988	+7.9% In review
INV-20362 · Pacific Substation Svcs \$127,600 vs est. \$121,400 · WO-117914	+5.1% In review
INV-20347 · Delta Line Constructors \$66,010 vs est. \$61,200 · WO-117802	+7.9% Resolved

INV-20391 · Delta Line Constructors
Work order WO-118427 · Received 7/14 · Cost engineering: J. Alvarez

Estimate \$84,200	Invoice \$99,350	Variance +\$15,150 (+18.0%)	Status Open
-----------------------------	----------------------------	---	-----------------------

Reason code * (managed list, owned by Finance)
Scope change: unplanned conduit replacement

Comments
Crew hit degraded conduit on segment 4. Replacement approved on site by inspector. PO amendment pending.

☒ **Approval required.** Variance exceeds the 10% threshold.
Routes to cost engineering in Teams - reminder after 48 hours

Submit with reason **Save draft**

Created 7/14 6:02 AM from SQL staging · Last change J. Short 7/16 9:12 AM · All changes logged

FIG. 1 · CANVAS APP · EXCEPTIONS QUEUE AND DISPOSITION FORM · FICTIONAL DATA

How it works

- 1 Exceptions surface from SQL staging.** Only invoices over the variance threshold reach the queue. The analyst investigates, then records the conclusion as a reason code and comment.
- 2 Reason codes are reference data owned by Finance.** The list lives in its own table, so codes change without an app release.
- 3 Variances over 10% start an approval.** The card lands in Teams, with a reminder at 48 hours and an alert to the owner if the flow errors.
- 4 Access is audited by design.** Security roles by team, column security on cost fields, every change logged in Dataverse.

WHY A CANVAS APP: THE NATIVE POWER BI VISUAL EMBEDS CANVAS APPS AND PASSES THE REPORT'S ROW SELECTION INTO THE APP. A MODEL-DRIVEN APP CAN BE IFRAMED INTO A REPORT, BUT WITHOUT THAT SELECTION CONTEXT. MODEL-DRIVEN STAYS THE RIGHT CALL FOR A CASE-MANAGEMENT BACK OFFICE ON THE SAME TABLES.

The operations and analytics hub

The Power Apps visual embeds the disposition form inside the Power BI report. The review reads the numbers and fixes the record on the same page.

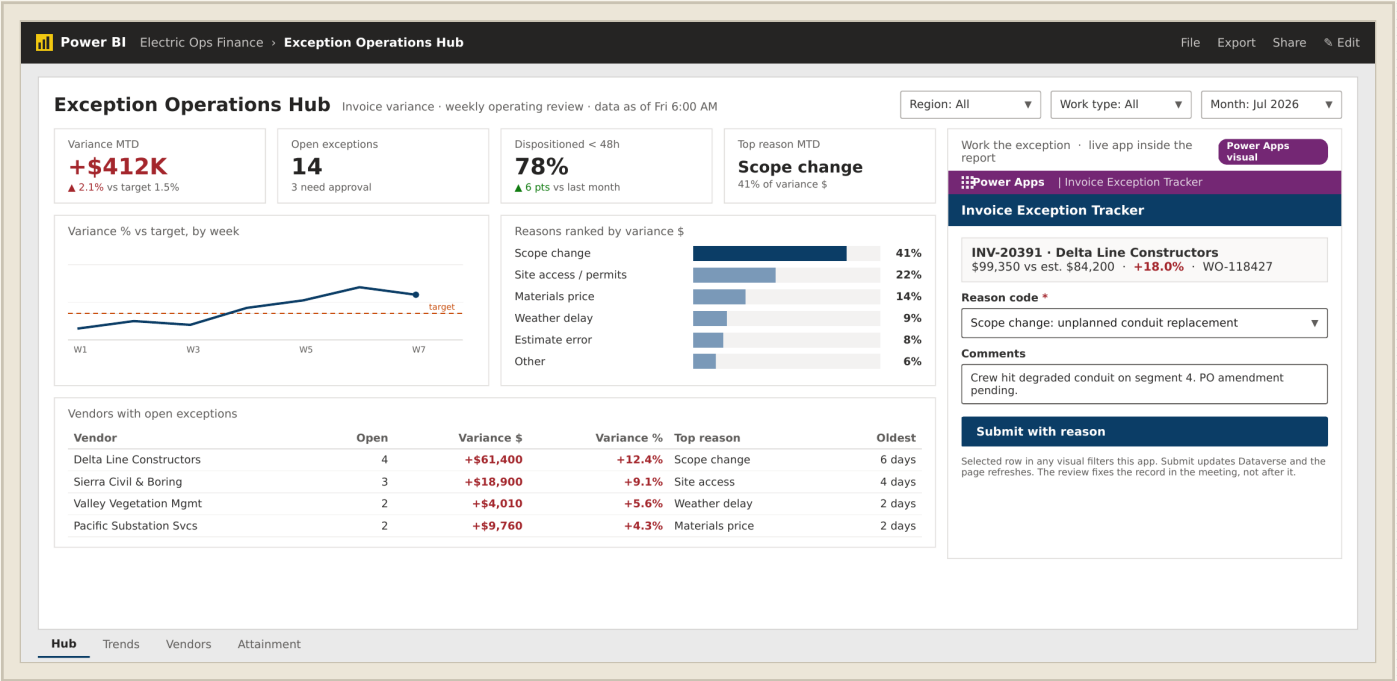


FIG. 2 · POWER BI REPORT WITH THE APP EMBEDDED AT RIGHT · FICTIONAL DATA

- **Click a vendor or invoice row and the embedded form opens on that record.** The selection carries into the app, so nobody re-searches the invoice.
- **Submit writes to Dataverse, then the app triggers a report refresh.** The exception shows resolved while the page is still open.
- **The report ranks what matters, the app records the decision.** Reasons ranked by variance dollars show which cause to attack first.
- **Slicers scope the meeting.** Region, work type, and month cut the queue to the room that is looking at it.

WHY EMBED THE APP AT ALL

Without embedding: the review spots INV-20391, someone writes it down, opens the app after the meeting, searches for the invoice, and the report shows it open until the next refresh. With embedding: the row is clicked, the form is already on that invoice, and the number moves before the meeting ends. No gap where follow-ups get lost.

The approval flow and the agent

The same Dataverse tables that run the app and the report feed the approval flow and the agent.

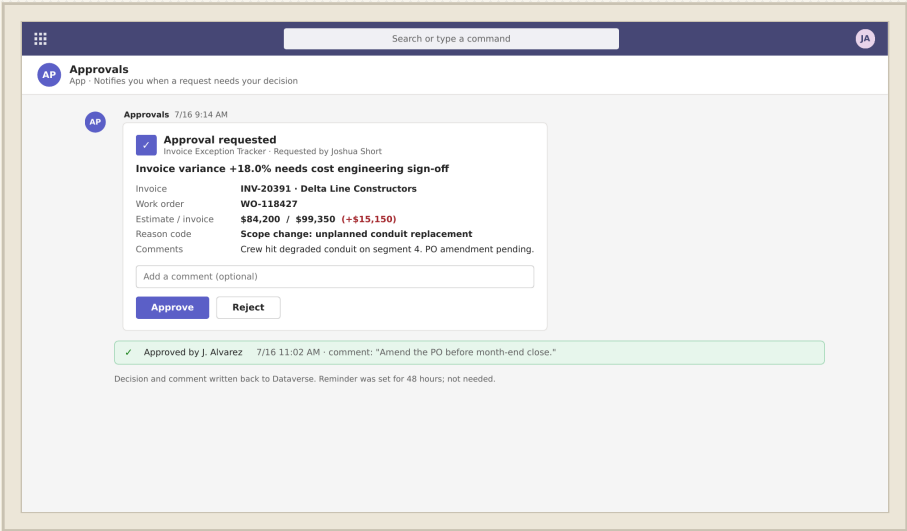


FIG. 3 · POWER AUTOMATE APPROVAL CARD IN TEAMS · THE DECISION WRITES BACK TO DATAVERSE

- **A Power Automate flow starts the approval when variance passes the threshold.** The approver decides in Teams; decision, comment, and timestamp write back to Dataverse.
- **Reminder at 48 hours.** If the flow errors, the owner is alerted the same hour. Unanswered approvals are how invoices age out.

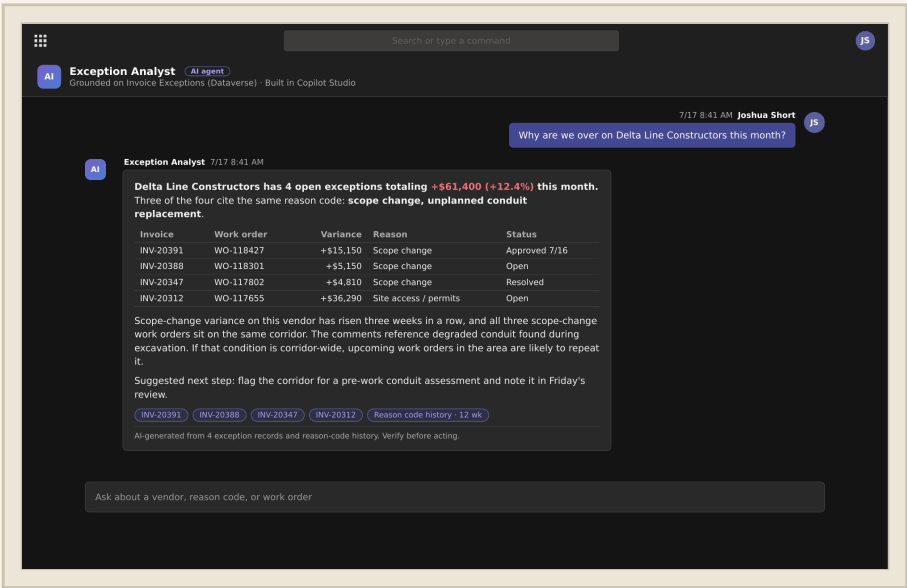


FIG. 4 · COPILOT STUDIO AGENT GROUNDED ON EXCEPTION HISTORY · EVERY ANSWER CITES ITS RECORDS

- **Ask about a vendor, get the pattern.** The agent reads exception history in Dataverse, cites the records behind every answer, and suggests the next step.
- **It ships scoped.** One knowledge source, a credit budget, DLP policies applied. A contained first agent that earns the second one.

Architecture

Dataverse holds the exceptions and the reason codes. The app, the report, the flow, and the agent are four surfaces on that one model.

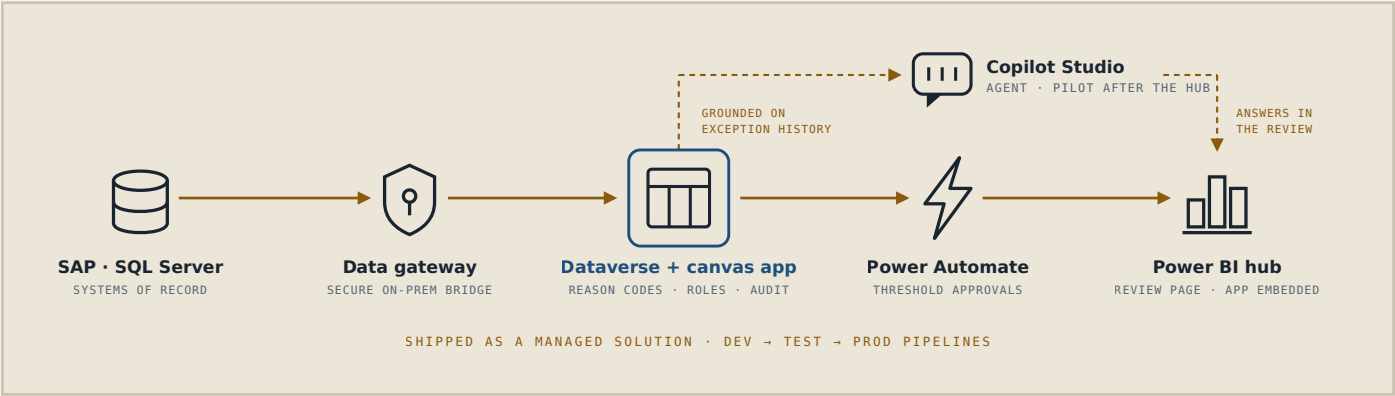


FIG. 5 · ONE DATAVERSE MODEL, FOUR SURFACES · THE AGENT SHIPS AS THE SECOND RELEASE

ASSUMPTIONS I WOULD VALIDATE IN WEEK ONE

- ? Where estimates and invoice status live today: the ERP directly, SQL staging, or both. This sketch assumes SQL Server in front of SAP.
- ? What an existing app already captures. If reason codes exist today, the win is the reference table, the embedded hub, and the write-back loop.
- ? Who owns reason codes and approval thresholds. This design keeps both in Finance's hands.

THE SAME DESIGN HANDLES FIELD WORK AS-IS: SWAP INVOICE EXCEPTIONS FOR JOB COMPLETIONS, KEEP THE REASON CODES, THE FLOW, AND THE HUB.